

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA1442 COURSE NAME: Compiler Design for High- Level Programming

COURSE OUTCOME COVERED

CO3: Analyze syntax-directed translation method using synthesized and inherited attributes to generate a Parser Tree. (BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 60 Total Marks:60

Annexure A

MCQ

Code of Conduct:

I V.Santhosh kumar /192411015(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.santjosh kumar Criteria Weightage Level 4 – Exemplary (50–60 Marks) Level 3 – Proficient (40–49 Marks) Level 2 – Developing (30–39 Marks) Level 1 – Beginning (0–29 Marks) Number of Correct Answers 100% 50 to 60 correct answers 40 to 49 correct answers 30 to 39 correct answers 0 to 29 correct answers Accuracy 100% Excellent (90–100%) Good (75–89%) Average (50–74%) Poor (Below 50%)

Course Faculty

Annexure A

MCQ 1. A compiler designer wants to attach semantic meaning to grammar productions in order to generate intermediate representations efficiently. Which compiler concept is most suitable for this purpose?

A) Lexical buffering B) Associating semantic rules with grammar productions C) Register allocation D) Peephole optimization

2. While compiling an arithmetic expression, a compiler internally creates a hierarchical representation showing operators and operands. Which structure is being generated?

A) Stack frame B) Queue structure C) Syntax Tree D) Linked List

3. A compiler evaluates attributes from child nodes before computing the parent node value during semantic analysis. Which type of attribute evaluation is taking place?

A) Parent-driven evaluation B) Synthesized attribute evaluation C) Inherited attribute evaluation D) Dynamic attribute evaluation

4. A language processor uses LR parsing to evaluate semantic information during reductions. Which attribute grammar is naturally suitable in this case?

A) Context-free attribute grammar

- B) Recursive attribute grammar
- C) S-attributed definition
- D) Circular attribute grammar

5. During semantic rule evaluation, a parser requires both parent information and child information while traversing grammar symbols from left to right. Which definition supports this requirement?

A) S-attributed definition B) Static semantic definition C) L-attributed definition D) Ambiguous definition

6. A compiler engineer wants child nodes to be processed before the parent node during semantic evaluation of expressions. Which traversal strategy should be selected?

A) Preorder traversal B) Inorder traversal C) Postorder traversal D) Level-order traversal

7. An expression such as $(a+b)*c$ is represented internally without unnecessary grammar details. What does this representation mainly capture?

A) Memory layout B) Hierarchical program structure C) Register allocation D) CPU execution order

8. While evaluating semantic rules, information is passed from a parent node toward its child nodes. Which type of attribute is involved?

- A) Synthesized attribute
- B) Terminal attribute
- C) Inherited attribute
- D) Dynamic attribute

9. A compiler rejects the statement `int x = "hello";` before execution begins. Which feature of the programming language is responsible for detecting this issue?

A) Token generation B) Type system C) Syntax optimization D) Loop unrolling

10. Consider the statement `float value = 25;`. Which compiler activity occurs automatically here?

A) Explicit casting B) Dynamic conversion C) Implicit type conversion D) Symbol replacement

11. A semantic analyzer verifies whether variables are declared and operations are type-compatible. Which compiler phase performs these tasks?

A) Lexical Analysis B) Syntax Analysis C) Semantic Analysis D) Linking

12. A compiler developer chooses a representation that eliminates unnecessary grammar symbols while preserving expression meaning. Which structure is preferred over a parse tree?

- A) Symbol Table
- B) Syntax Tree
- C) Token Stream
- D) Finite Automaton

13. During compilation, all information regarding identifiers, scope, and datatype is maintained centrally. Which compiler component performs this responsibility?

A) Code Generator B) Symbol Table C) Syntax Parser D) Optimizer

14. A compiler uses only synthesized information for semantic evaluation of grammar productions. Which type of attribute grammar is being implemented?
A) Dynamic grammar B) L-attributed grammar C) S-attributed grammar D) Context-sensitive grammar
15. A recursive descent parser performs semantic translation while expanding productions from the start symbol downward. Which translation approach is being followed?
A) Bottom-up translation B) Top-down translation C) Shift-reduce translation D) Operator precedence translation
16. During shift-reduce parsing, intermediate grammar symbols are temporarily stored before reductions occur. Which data structure supports this operation most effectively?
A) Queue
B) Heap
C) Stack
D) Linked List
17. A compiler reports an error when a boolean value is added to an integer variable. What compiler mechanism detected this problem?
A) Syntax checking B) Type checking C) Token validation D) Memory management
18. A programmer declares a variable to store whole numbers without decimal values. Which datatype is most appropriate?
A) Boolean B) Integer C) Character D) String
19. Two array types are considered equivalent because they possess identical internal structures even though their names differ. Which type equivalence principle is applied?
A) Name equivalence B) Structural equivalence C) Functional equivalence D) Dynamic equivalence
20. A compiler implementing LR parsing evaluates semantic rules during reductions. Which evaluation strategy is best suited for this implementation?
A) Top-down evaluation
B) Parallel evaluation
C) Bottom-up evaluation
D) Recursive evaluation
21. During expression analysis, a compiler stores operators as internal nodes and operands as leaves. Which representation is being used?
A) AVL Tree B) Heap Tree C) Syntax Tree D) B-Tree
22. A semantic rule requires information from symbols appearing to the left side of a production. Which attribute grammar supports such evaluation?
A) S-attributed grammar B) L-attributed grammar C) Ambiguous grammar D) Recursive grammar

23. After semantic analysis, a compiler converts validated source constructs into machine-independent instructions. Which compiler phase performs this task?

A) Lexical Analysis B) Intermediate Code Generation C) Linking D) Parsing

24. A programmer intentionally converts a floating-point value into an integer using a language construct. What is this process called?

- A) Promotion
- B) Reduction
- C) Casting
- D) Coercion

25. While scanning source code, a compiler groups characters into meaningful units such as keywords and identifiers. Which phase performs this operation?

A) Semantic Analysis B) Syntax Analysis C) Lexical Analysis D) Optimization

26. A parse tree generated for a program statement explicitly contains all grammar symbols used in derivation. What is the primary characteristic of this structure?

A) Contains machine instructions B) Represents grammar symbols C) Removes terminals D) Stores registers only

27. A bottom-up parser starts constructing derivations beginning with identifiers and constants before reaching the start symbol. Which nodes are processed first?

A) Root nodes B) Internal nodes C) Leaf nodes D) Header nodes

28. During assignment checking, a compiler ensures the datatype of the expression matches the datatype of the variable receiving it. Which component performs this verification?

- A) Loader
- B) Semantic checker
- C) Tokenizer
- D) Syntax formatter

29. Two variables are treated as equivalent only because they were declared using the same type name. Which equivalence mechanism is being followed?

A) Structural equivalence B) Dynamic equivalence C) Name equivalence D) Functional equivalence

30. A semantic evaluator computes all child expressions before processing the parent expression. Which traversal technique reflects this behavior?

A) Preorder traversal B) Inorder traversal C) Postorder traversal D) Reverse traversal

31. A compiler designer inserts executable semantic instructions directly inside grammar productions to perform translation tasks. What is this mechanism called?

A) Syntax-directed translation scheme B) Token generation mechanism C) Recursive grammar scheme D) DFA representation

32. A parser repeatedly performs shift and reduce operations until the start symbol is produced. Which parsing strategy is being used?

- A) Predictive parsing
- B) Top-down parsing
- C) Bottom-up parsing
- D) Recursive parsing

33.A strongly typed programming language prevents invalid arithmetic operations between incompatible datatypes. Which language feature enforces this?

- A) Type system B) Code optimizer C) Symbol allocator D) Loader

34.A programmer creates a collection capable of storing multiple values of the same datatype under one name. Which datatype category does this belong to?

- A) Primitive datatype B) Derived datatype C) Boolean datatype D) Character datatype

35.In a syntax tree representing an assignment statement, the highest node corresponds to the complete statement meaning. Which element does the root node generally represent?

- A) Constant value B) Identifier name C) Entire expression or statement D) Memory address

36.A compiler reports an error because a variable is used outside its declared scope. Which compiler phase detects this issue?

- A) Syntax Analysis
- B) Semantic Analysis
- C) Code Optimization
- D) Linking

37.A compiler automatically converts an integer operand into a floating-point operand before arithmetic evaluation. What type of conversion is this?

- A) Narrowing conversion B) Implicit widening conversion C) Explicit casting D) Recursive conversion

38.During attribute evaluation, semantic rules are processed from left to right so that inherited information becomes available when required. Which attribute grammar follows this restriction?

- A) Circular grammar B) L-attributed grammar C) S-attributed grammar D) Ambiguous grammar

39.A compiler uses shift-reduce actions and reduction tables to parse complex programming constructs. Which parser is most likely being used?

- A) Predictive parser B) Recursive descent parser C) LR parser D) LL(1) parser

40.A compiler engineer associates semantic computations directly with grammar rules for translation purposes. To what are these semantic rules attached?

- A) Tokens only
- B) Grammar productions
- C) Machine registers
- D) Memory blocks

41.During semantic evaluation, a node computes its value entirely using values obtained from child nodes. Which attribute type is involved?

- A) Inherited attribute B) Dynamic attribute C) Synthesized attribute D) External attribute

42. A compiler designer prefers a compact internal representation that removes unnecessary non-terminal symbols from derivations. Which representation is best suited?

A) Parse Tree B) Syntax Tree C) NFA D) Transition Graph

43. A program contains the statement `total = amount + tax;` where `tax` has never been declared. Which compiler phase identifies this error?

A) Lexical Analysis B) Parsing C) Semantic Analysis D) Code Generation

44. A compiler automatically changes the datatype of an operand during expression evaluation to avoid type mismatch. What is this process called?

- A) Coercion
- B) Parsing
- C) Reduction
- D) Folding

45. Two structures are treated as equivalent because they contain identical field arrangements and datatypes. Which equivalence principle is being applied?

A) Name equivalence B) Scope equivalence C) Structural equivalence D) Dynamic equivalence

46. A compiler implementing LR parsing evaluates semantic actions during reduce operations. Which evaluation strategy aligns best with this parser behavior?

A) Predictive evaluation B) Recursive evaluation C) Bottom-up evaluation D) Top-down evaluation

47. A compiler developer chooses an attribute grammar that can be evaluated naturally during reduce actions without inherited information. Which grammar type is most appropriate?

A) L-attributed grammar B) S-attributed grammar C) Ambiguous grammar D) Circular grammar

48. In a syntax tree for arithmetic expressions, the terminal values such as variables and constants appear at which location?

- A) Internal nodes
- B) Root node
- C) Leaf nodes
- D) Header nodes

49. A programmer explicitly converts a floating-point result into an integer before storing it in a variable. Which operation is being applied?

A) Promotion B) Casting C) Translation D) Coercion

50. After successful syntax checking, a compiler begins verifying meanings, datatypes, and scope rules. Which phase starts next?

A) Lexical Analysis B) Semantic Analysis C) Linking D) Loading

51. A programming language verifies datatype compatibility during compilation and sometimes again during execution. What does this indicate about type checking?

A) It occurs only during loading B) It occurs only during runtime C) It can occur during compile time and runtime D) It is unnecessary in modern languages

52.A parser constructs derivations beginning from tokens and gradually reduces them until the start symbol is obtained. Which parsing approach is demonstrated?

- A) Top-down parsing
- B) Predictive parsing
- C) Bottom-up parsing
- D) Recursive parsing

53.A compiler internally represents expressions using parent-child relationships between operators and operands instead of linear text order. Which representation style is being followed?

- A) Sequential representation
- B) Hierarchical representation
- C) Tabular representation
- D) Circular representation

54.A compiler phase determines whether expressions, assignments, and declarations carry valid meaning according to language rules. Which aspect is primarily analyzed?

- A) Syntax structure
- B) Meaning of constructs
- C) Register allocation
- D) Instruction scheduling

55.Two datatype declarations are accepted as compatible because the language rules consider them logically interchangeable. What does this describe?

- A) Operator precedence
- B) Variable scope
- C) Type equivalence
- D) Memory allocation

Answers: 1. B) Associating semantic rules with grammar productions 2. C) Syntax Tree 3. B) Synthesized attribute evaluation 4. C) S-attributed definition 5. C) L-attributed definition 6. C) Postorder traversal 7. B) Hierarchical program structure 8. C) Inherited attribute 9. B) Type system 10.C) Implicit type conversion 11.C) Semantic Analysis 12.B) Syntax Tree 13.B) Symbol Table 14.C) S-attributed grammar 15.B) Top-down translation 16.C) Stack 17.B) Type checking 18.B) Integer 19.B) Structural equivalence 20.C) Bottom-up evaluation 21.C) Syntax Tree 22.B) L-attributed grammar 23.B) Intermediate Code Generation 24.C) Casting 25.C) Lexical Analysis 26.B) Represents grammar symbols 27.C) Leaf nodes 28.B) Semantic checker 29.C) Name equivalence 30.C) Postorder traversal 31.A) Syntax-directed translation scheme 32.C) Bottom-up parsing 33.A) Type system 34.B) Derived datatype 35.C) Entire expression or statement

36.B) Semantic Analysis 37.B) Implicit widening conversion 38.B) L-attributed grammar 39.C) LR parser 40.B) Grammar productions 41.C) Synthesized attribute 42.B) Syntax Tree 43.C) Semantic Analysis 44.A) Coercion 45.C) Structural equivalence 46.C) Bottom-up evaluation 47. B) S-attributed grammar 48.C) Leaf nodes 49.B) Casting 50.B) Semantic Analysis 51.C) It can occur during compile time and runtime 52.C) Bottom-up parsing 53.B) Hierarchical representation 54.B) Meaning of constructs 55.C) Type equivalence

RUBRICS FOR CALCULATION

Criteria Weightage Level 4 – Exemplary (90–100 Marks) Level 3 – Proficient (75–89 Marks) Level 2 – Developing (50–74 Marks) Level 1 – Beginning (0–49 Marks) Number of Correct Answers 100% 90 to 100 correct answers 75 to 89 correct answers 50 to 74 correct answers 0 to 49 correct answers

Criteria Weight age (%) Level 4 – Exemplary

(5 Marks) Level 3 – Proficient

(4 Marks) Level 2 – Developing (2–3 Marks) Level 1 – Beginning (0–1 Mark) Conceptual

Understanding 30 Demonstrates clear and complete understanding of the concept Good understanding with minor gaps Basic understanding with some misconceptions Poor or incorrect understanding Accuracy of Answer 25 Completely correct answer with no errors Mostly correct with minor errors Partially correct with noticeable errors Incorrect or irrelevant answer Application of Knowledge 20 Correctly applies concepts to solve the question/problem Applies concepts with minor mistakes Limited or partially correct application No or incorrect application Completeness of Response 15 Covers all required parts of the question Covers most parts with minor omissions Covers only some parts Incomplete or missing response Clarity & Presentation 10 Clear, concise, and Understandable with Somewhat unclear or Difficult to understand

Criteria Weight age (%) Level 4 – Exemplary

(5 Marks) Level 3 – Proficient

(4 Marks) Level 2 – Developing (2–3 Marks) Level 1 – Beginning (0–1 Mark) well-organized answer minor issues poorly structured